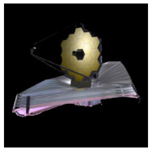

7901 - Improving JWST/NIRSpec Data Reduction: Addressing Correlated Noise and 1D Spectral Extraction Limitations

Cycle: 4, Proposal Category: AR

INVESTIGATORS

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OBSERVATIONS

ABSTRACT

This proposal aims to integrate JWST NIRSpec data reduction capabilities into the PypeIt spectroscopic data reduction package, in order to address critical limitations in the current JWST calwebb_spec3 pipeline. The standard approach in the calwebb_spec3 pipeline of resampling dithered spectra into a regular grid introduces correlation between pixels that reduce the accuracy of noise estimates, affecting particularly the detection of faint sources. Additionally, the lack of optimal extraction tools in the existing JWST pipeline has necessitated the use of custom codes, which creates inconsistencies in data analysis.

PypeIt offers a solution by providing native pixel sampling using an irregular wavelength grid, robust noise estimates, and both optimal and boxcar 1D extraction. PypeIt has been successfully used to reduce data from over 50 spectrographs at 18 observatories and is recognized as a valuable community resource. By extending PypeIt to support JWST NIRSpec fixed-slit and micro-shutter assembly observations, we will provide the JWST community with access to highly vetted, well-maintained, open-source algorithms developed over decades. Our proof-of-concept work demonstrates PypeIt's potential to improve JWST data reduction, particularly in noise estimation and spectral extraction.

This project will develop automated workflows for various NIRSpec observations, optimize existing code, create new output formats and quality assessment tools, and produce extensive documentation. The integration of JWST capabilities into PypeIt will expedite the path to science, improve result reproducibility, and facilitate comparisons between datasets from different observatories.

OBSERVING DESCRIPTION